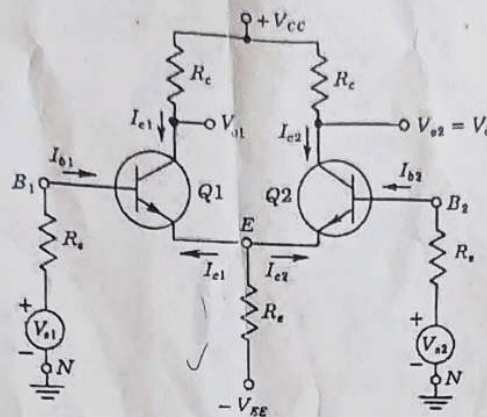
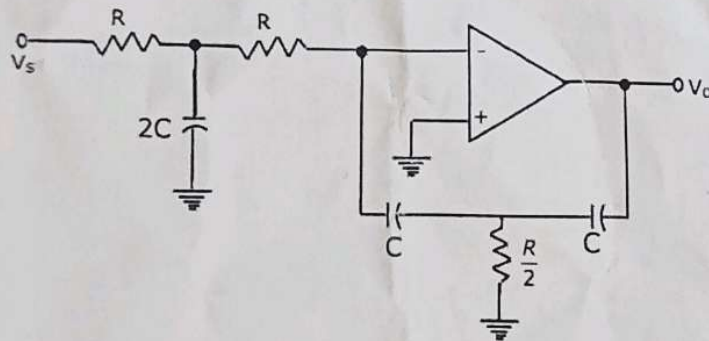


Note: Attempt all

1. Assume that $R_s = 0, h_{oe}(R_c + 2R_e) \ll 1, h_{fe} \gg 1$ and $h_{ie} \ll 2R_e h_{fe}$ in the circuit shown in the Figure below. i) Verify that the CMRR is given by $\rho = \frac{h_{fe} R_e}{h_{ie}}$
 ii) Verify that $\rho = g_m R_e \approx \frac{V}{2V_T}$ for $r_{bb'} \ll r_{b'e}$. Here, V is the quiescent voltage across R_c

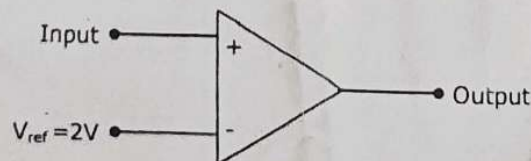


2. The system shown in the Figure below is a double integrator. Find the transfer gain $(V_o(s)/V_i(s))$, assume ideal OP-AMP.

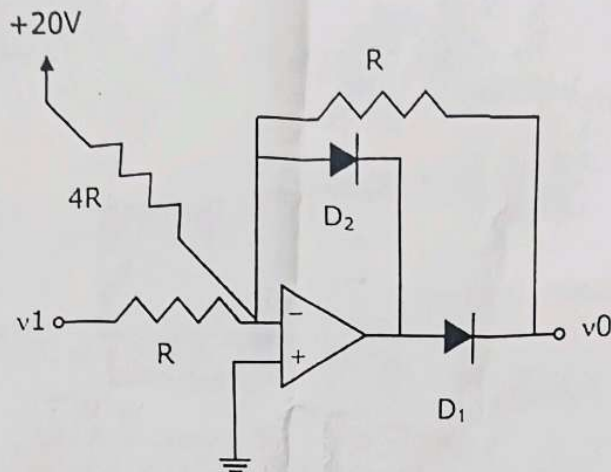


OR

If the input to the ideal comparator shown in figure is sinusoidal signal of 8V (peak to peak) without any DC component, then find a duty cycle of the output of the comparator?



3. The transfer function of an OPAMP has its first pole at 1 MHz and a low frequency gain of 44 dB. Dominant pole compensation is used for this OPAMP, and the gain of the compensated amplifier is zero dB at a frequency of 1 MHz. Find the value f_d of the dominant pole. Repeat the problem if the low frequency gain is 68 dB.
4. Find the transfer characteristic for the precision rectifier circuit shown below is (assume ideal OP-AMP and practical diodes)



OR

Design a 4th-order Butterworth low-pass filter with cutoff frequency 2 kHz and pass band gain is 1